

THE ROOTED-TREE NORMAL FORM: DIRECTED CONTAINERS, FREE MONOIDS, AND ARBOREAL GAME COMONADS

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ABSTRACT. A recurring move in the theory of containers and polynomial functors replaces an *arbitrary-position* condition by a *rooted, tree-shaped* discipline, under which the operative backward map reads only a unique root-path. We isolate this move and show that it is the same move in four independently-developed settings: the root axiom that turns a container into a comonad (Ahman–Uustalu), the path-confinement of the backward pass of a branching free monoid, the tree-connectedness axiom that repairs the theory of arboreal categories (Jabl–Reggio), and the forest condition on polynomial comonoids. The move has a single geometric content—in a forest the path to the root is unique—which in the last and sharpest instance becomes a theorem: the Eilenberg–Moore coalgebras of a polynomial comonad $c = C$ are the copresheaves on C , so the coalgebra category is arboreal exactly when C is a forest. This identifies the arboreal coalgebra categories of finite model theory with copresheaf categories of polynomial comonads, giving a bridge between the game-comonad programme and the Poly programme. We close with an honestly-graded open problem on how the two meet at the level of fibrations.

1. INTRODUCTION

A move one keeps making. Here is a pattern. One is handed a construction—a container, a free monoid, a category of coalgebras—together with a condition phrased over *all* positions, or *all* coproducts, or *all* arrows. The condition looks hard to control because the positions branch: different answers lead to different continuations, different subobjects combine in uncontrolled ways. Then one notices that the condition can be re-phrased so that the data is confined to a *rooted tree*, and that the map one actually has to understand reads only a *single root-path*—a unique prefix, a unique minimal element. The branching that threatened to create an obstruction is forbidden by the tree shape, and the obstruction evaporates.

This paper is about that move. We have met it four times, in four settings developed independently of one another, by different communities, for different reasons. The purpose of the paper is to state the move once, precisely, and to exhibit the four settings as instances of it—so that the

reader who knows one of them recognises the others, and so that a tool sharpened in one can be carried to the rest.

The four settings. The four instances, in order of increasing categorical sharpness, are these.

- (I) *The root axiom.* A container $(S \triangleleft P)$ is a comonad on $\mathbf{Set}$ —equivalently a small category—exactly when its positions carry a distinguished root at each shape together with a compatible codomain-and-composition discipline. This is the founding instance: the move is already the definition of a *directed* container [3, 4].
- (II) *The prefix cut.* The free monoid on a container C under composition—a type of answer-driven interaction tree—carries a multiplication whose forward part grafts trees and whose backward part splits a leaf-path. We prove (Lemma 4.1) that the backward part is *independent of the grafted data*: it cuts a leaf-path at its unique prefix that is a leaf of the outer tree. The feared obstruction to a bidirectional universal property—that answer-dependence corrupts the backward pass—cannot occur, because the cut sees only the outer tree.
- (III) *Factoring through the least element.* In the theory of *arboreal categories* [2, 7], an axiom requiring paths to be *connected* fails for the modal game comonads. Jakl and Reggio repair it with *tree-connectedness*: an arrow into a colimit of a tree-diagram of paths must factor through a *least* element of the tree [7, Def. 11]. This is the same move—the factoring is through a unique minimal element—and it recovers all the pebbling and Ehrenfeucht–Fraïssé results while admitting the modal comonads.
- (IV) *The base is a forest.* The move becomes a property of a small category. We show (Lemma 4.5) that the Eilenberg–Moore coalgebras of a polynomial comonad $c = C$ are the copresheaves $[C, \mathbf{Set}]$; for the Ehrenfeucht–Fraïssé comonad this is the category of forests of bounded height. Hence (Corollary 4.6) “*the coalgebra category is arboreal*” is exactly “*C is a forest*”. The rooted-tree discipline is now a property of the base category, and arboreality is that property read through the equivalence *polynomial comonad* = *directed container* = *small category*.

The main result, and what is and is not new. The load-bearing theorem is instance (IV):

Theorem 1.1 (see Lemma 4.5 and Corollary 4.6). *Let $c = C$ be a polynomial comonad, viewed as a small category. The category of Eilenberg–Moore coalgebras of the comonad $\llbracket c \rrbracket$ on $\mathbf{Set}$ is equivalent to the copresheaf category $[C, \mathbf{Set}]$, i.e. to the discrete opfibrations over C . When C is a forest category these copresheaves are forests; for the Ehrenfeucht–Fraïssé comonad, whose comonoid is the linear order $[k]$, they are exactly the forests of height $\leq k$ of finite model theory, and for the pebbling comonad the trie-indexed*

forests. The arboreal coalgebra categories of these game comonads are thus the copresheaf categories of their polynomial comonoids.

We are careful about novelty. That the coalgebras of a polynomial comonad are copresheaves on its category is *folklore-adjacent* in the Poly community—it is the familial / cofree-coalgebra story [8]—and that the Ehrenfeucht–Fraïssé coalgebras are forests is proved directly by Abramsky and Shah [2]. What we contribute is not a new deep theorem but the *clean identification tying the two*: the one-line reason that turns “arboreality = the base is a forest” from an analogy into a statement with a proof, and the observation that this is the fourth and sharpest appearance of a single structural move. The value here is the *bridge*—between the game-comonad programme of finite model theory and the Poly programme of compositional systems—not a priority claim.

Why a reader should care. Two reasons. First, pedagogical: once one sees that the root axiom, the prefix cut, the tree-connectedness repair, and the forest condition are one move, each becomes easier to remember and to apply, and a technique from one setting (for instance, the path-confinement argument of Section 4.2) suggests itself in the others. Second, structural: Theorem 1.1 says the arboreal coalgebra categories that finite model theory studies are exactly copresheaf categories of polynomial comonads. The two programmes have never cited one another on this point; the identification lets results and intuitions cross. We make the bridge precise, and in Section 5 we push it to the level of fibrations, where it becomes a genuine, and genuinely open, question—stated there with its grade and its obstructions named.

Outline. Section 2 fixes containers, the composition product, the equivalence of polynomial comonoids with small categories, directed containers, and forests. Section 3 states the normal form once, as a schema, after isolating its single geometric content. Section 4 walks the four instances in order of increasing sharpness, proving instances (II) and (IV) and citing instances (I) and (III). Section 5 reads the four as one and states the open problem that the schema opens, honestly graded. Section 6 concludes.

Provenance. Instances (I) and (III) are established and cited; instances (II) and (IV) are proved here, the former by structural induction with a computational check, the latter by a direct argument. The material of Section 5 beyond the stated theorem is explicitly graded *computed* or *open*. We have tried to keep the claims exactly as strong as the proofs behind them, and no stronger.

2. PRELIMINARIES

We recall only what the four instances use. Throughout, $\mathbf{Set}$ is the category of sets.

2.1. Containers and polynomial functors. A *container* $C = (S \triangleleft P)$ is a set S of *shapes* together with a family $P : S \rightarrow \mathbf{Set}$ of *positions*. Its *extension* is the functor

$$\llbracket C \rrbracket : \mathbf{Set} \rightarrow \mathbf{Set}, \quad \llbracket C \rrbracket A = \sum_{s \in S} A^{P_s},$$

a coproduct of representables; functors of this form are the *polynomial* functors. A morphism of containers $\varphi : (A \triangleleft P_A) \rightarrow (B \triangleleft P_B)$ is a *forward* shape map $\varphi_1 : A \rightarrow B$ together with a *backward* family $\varphi_a^\sharp : P_B(\varphi_1 a) \rightarrow P_A(a)$, contravariant in positions. Containers and their morphisms form a category $\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Set}^{\text{op}})$, often written $\mathbf{Poly}$. We use the two freely.

The reading that organises instance (II) is the *bidirectional* one: a shape $s \in S$ is a request, and a position $p \in P_s$ is a possible answer to it.

Proposition 2.1 (Gambino–Kock [5], Thm. 4.5). *The extension functor $\llbracket - \rrbracket : \mathbf{Poly} \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is fully faithful: for containers p, q ,*

$$\mathbf{Poly}(p, q) \cong \mathbf{Nat}(\llbracket p \rrbracket, \llbracket q \rrbracket).$$

2.2. The composition product and polynomial comonoids. The *composition product* $\triangleleft$ on $\mathbf{Poly}$ computes functor composition, $\llbracket p \triangleleft q \rrbracket = \llbracket p \rrbracket \circ \llbracket q \rrbracket$, with unit $y = (1 \triangleleft 1)$, the identity functor. Concretely,

$$(G \triangleleft F).\text{Shape} = \sum_{t \in T} (Qt \rightarrow S), \quad (G \triangleleft F).\text{Pos}(t, f) = \sum_{q \in Qt} P(fq).$$

A *comonoid* in the monoidal category $(\mathbf{Poly}, \triangleleft, y)$ is a container c with a counit $\varepsilon : c \rightarrow y$ and a comultiplication $\delta : c \rightarrow c \triangleleft c$ satisfying the comonoid laws.

Proposition 2.2 (Ahman–Uustalu [4]; Spivak–Garner–Fairbanks [8]). *Comonoids in $(\mathbf{Poly}, \triangleleft, y)$ are precisely small categories. Under the equivalence, a comonoid $c = (S \triangleleft P)$ has objects S and arrows-out-of- s the positions P_s ; the counit picks, at each s , an identity arrow $\text{id}_s \in P_s$; the comultiplication provides, for each s , a codomain map $\text{cod}_s : P_s \rightarrow S$ and a composition $\sum_{f \in P_s} P(\text{cod}_s f) \rightarrow P_s$, $(f, g) \mapsto g \circ f$. The comonoid laws are the category axioms.*

Combining Propositions 2.1 and 2.2: a comonad on $\mathbf{Set}$ whose underlying functor is polynomial is *automatically* a polynomial comonoid (the counit and comultiplication, being natural transformations between polynomial functors, are extensions of unique container morphisms by full faithfulness, and the comonad laws transport), hence a small category. We write $c = C$ for a polynomial comonad and the small category it is.

2.3. Directed containers. A *directed container* [3, 4] is exactly such a structure presented element-wise: a container $(S \triangleleft P)$ with, for each shape s , a *root* position $o_s \in P_s$, a *subshape* $s \downarrow p \in S$ for each $p \in P_s$, and an addition $p \oplus q \in P_s$ for $q \in P(s \downarrow p)$, satisfying unit and associativity laws. These are the category axioms of Proposition 2.2, with $o_s = \text{id}_s$, $s \downarrow p = \text{cod}_s p$, and $\oplus$ composition. We write $\mathbf{DCont} \cong \mathbf{Cat}$ for the resulting equivalence.

2.4. Forests, paths, and copresheaves. A poset is a *forest* if the down-set $\{x : x \leq y\}$ of every element y is a finite chain; a *tree* is a forest with a least element. We view a small category C as generating a reachability preorder; C is a *forest category* when that preorder is a forest, so that from every object there is a *unique* directed path to a root. The *linear order* $[k] = (\{1, \dots, k\}, \leq)$ and the k -ary *prefix trie* on nonempty words $[k]^+$ are forests.

For a small category C , a *copresheaf* is a functor $X : C \rightarrow \mathbf{Set}$; the copresheaf category $[C, \mathbf{Set}]$ is equivalently the category of *discrete opfibrations* over C . This is the shape in which instance (IV) is stated.

3. THE NORMAL FORM

3.1. The single geometric fact. All four instances draw on one fact about forests, which we isolate before stating the schema.

Lemma 3.1 (Root-path uniqueness). *Let F be a forest. The following coincide for an element $y \in F$:*

- (1) *the down-set $\{x : x \leq y\}$, which is a finite chain;*
- (2) *the unique directed path from y to its root;*
- (3) *the unique minimal (equivalently least) element of that down-set;*
- (4) *for the tree of prefixes of a path ending at y , the unique prefix lying in any fixed prefix-closed subset.*

In each guise the data attached to y “below” it is a chain, carrying no branching.

Proof. A forest is exactly a poset in which every principal down-set is a chain, which is the equivalence of (1) with the uniqueness of the descending path (2); a nonempty finite chain has a least element, giving (3); and in a well-founded tree the leaves form a prefix code—no leaf is a proper prefix of another—so the prefix of a given path lying in a prefix-closed set is unique, giving (4). The last clause restates that a chain does not branch. $\square$

The four instances differ only in *which* guise of Lemma 3.1 they use: instance (I) distinguishes the root (3); instance (II) cuts at the unique prefix (4); instance (III) factors through the least element (3); instance (IV) makes “every down-set is a chain” (1) a property of the base category.

3.2. The schema. We can now say what the move is.

Schema 3.2 (The rooted-tree normal form). A condition phrased over *arbitrary* positions, coproducts, or arrows is replaced by a *rooted, tree-shaped* discipline: the data is confined to a forest, and the operative backward or factoring map is determined by the *unique root-path* of Lemma 3.1—a unique prefix, a least element, a chain down-set. Because a chain does not branch, the branching data that would have obstructed the condition is never read, and the condition holds.

Schema 3.2 is not itself a theorem—it is the shape that the four theorems of the next section share. In instance (IV) it becomes a theorem verbatim: the condition (arboreality) is *equivalent* to the forest property of the base. We take that equivalence as the evidence that the schema is real and not merely a family resemblance.

4. THE FOUR INSTANCES

4.1. Instance (I): positions carry a root. The founding instance is the definition of a directed container. A bare container $(S \triangleleft P)$ has *arbitrary* positions; there is no way to compose an observation at s with an observation at a position of s , because a position carries no shape. The directed-container structure supplies exactly the missing rooted discipline: a root $o_s \in Ps$ to serve as identity, a subshape $s \downarrow p$ to serve as codomain, and an addition to serve as composition (Proposition 2.2). This is the move at its most elementary—guise (3) of Lemma 3.1, the distinguished least element—and it is *necessary*: a container is a comonad if and only if it admits such a structure [3], i.e. if and only if it is a small category [4]. We record it as the base case and move on.

4.2. Instance (II): the prefix cut. The second instance is the backward pass of a free monoid, and it is where the move does real work.

4.2.1. The branching free monoid. Read a container $C = (S \triangleleft P)$ bidirectionally: S requests, Ps answers. Then the composition product $C \triangleleft D$ is *answer-dependent sequencing*—a first request s together with, for each possible answer $p \in Ps$, a second request—and the free monoid on C under $\triangleleft$ is the type of answer-driven interaction trees over C . This gadget is Hedges’ “Kleene star on containers” [6], which rebuilds a bidirectional typechecker on it but proves neither its monoid laws nor a universal property.

Concretely [5], the free monoid is $C^* = (S^* \triangleleft P^*)$ where S^* is closed P -trees $t ::= \text{lf} \mid \text{nd } s \kappa$ with $s \in S$ and $\kappa : Ps \rightarrow S^*$, and $P^*t = \text{leaves}(t)$ is the set of root-to-lf paths, recorded as the tuple of positions taken. A leaf is thus a *complete answer-history* of one run. The unit is lf; the multiplication grafts and splits,

$$\begin{aligned} \text{graft}(\text{lf}, u) &= u(()), & \text{graft}(\text{nd } s \kappa, u) &= \text{nd } s (\lambda p. \text{graft}(\kappa_p, u_p)), \\ \text{split}(\text{lf}, u, z) &= \langle (), z \rangle, & \text{split}(\text{nd } s \kappa, u, \langle p, z \rangle) &= \langle \langle p, \ell \rangle, w \rangle, \end{aligned}$$

where in the last line $(\ell, w) = \text{split}(\kappa_p, u_p, z)$ and $u_p := \lambda r. u \langle p, r \rangle$. The three monoid laws of $(C^*, \text{graft}, \text{lf})$ hold for every container; this is established elsewhere (the universal property of the free monad on a container), and the underlying $\triangleleft$ -monoid laws are formalised in Lean. The backward pass of the multiplication is split.

4.2.2. *Path-confinement.* The worry Hedges' construction raises is the bidirectional one: since the continuation after a request branches on the *answer*, does that answer-dependence corrupt the backward pass split, so that it fails to be natural in the grafted family and no universal property can hold? The following lemma says it cannot, and is the content of the move in this setting.

Lemma 4.1 (Path-confinement). *For every container C , every tree $t \in S^*$, every family $u : \text{leaves}(t) \rightarrow S^*$, and every leaf $z \in \text{leaves}(\text{graft}(t, u))$,*

$$\text{split}(t, u, z) = (\ell, w),$$

where ℓ is the unique prefix of z that is a leaf of t and w is the corresponding suffix. In particular $\text{split}(t, u, z)$ is independent of u : the backward pass of the multiplication reads only the outer tree.

Proof. Structural induction on t , using guise (4) of Lemma 3.1.

Base $t = \text{lf}$. By definition $\text{split}(\text{lf}, u, z) = \langle (), z \rangle$. The empty path $()$ is a prefix of every z and is the unique leaf of lf , so $\ell = ()$, $w = z$; the output did not consult u .

Step $t = \text{nd } s \kappa$. A leaf of $\text{graft}(\text{nd } s \kappa, u) = \text{nd } s(\lambda p. \text{graft}(\kappa_p, u_p))$ has the form $z = \langle p, z' \rangle$ with $p \in Ps$ and $z' \in \text{leaves}(\text{graft}(\kappa_p, u_p))$. By definition $\text{split}(\text{nd } s \kappa, u, z) = \langle \langle p, \ell' \rangle, w' \rangle$ with $(\ell', w') = \text{split}(\kappa_p, u_p, z')$. By the induction hypothesis applied to the subtree κ_p with family u_p , the pair (ℓ', w') depends only on κ_p and z' , and ℓ' is the unique κ_p -leaf prefix of z' . Now a prefix of $z = \langle p, z' \rangle$ is a leaf of $\text{nd } s \kappa$ iff it has the form $\langle p, r \rangle$ with r a leaf-prefix of z' in κ_p ; such r exists and is unique, namely $r = \ell'$. Hence $\langle p, \ell' \rangle$ is the unique $(\text{nd } s \kappa)$ -leaf prefix of z , with suffix w' , and the computation used only t and z . $\square$

Corollary 4.2. *The answer-dependence of a container's continuations is confined to the shape data of C^* (the trees and their per-answer children) and to the leaves (the complete answer-histories); it never enters the backward pass of the monoid multiplication. The one-directional proof of the universal property of C^* is therefore already the bidirectional proof, and the feared obstruction does not exist.*

Proof. The backward component of associativity is, by Lemma 4.1, a statement purely about paths: both split-composites of the associator read only their respective outer trees, so the double split of a leaf of $\text{graft}(\text{graft}(t, u), v)$ is the unique three-fold factorisation into outer-leaf, middle-leaf, and inner suffix, whose well-definedness is associativity of path concatenation; no answer datum from u or v is read. The unit laws are trivial. Hence no monoid law of C^* consults the grafted data in its backward component. $\square$

Remark 4.3 (The linear case is O'Neill). When S is a singleton the trees are lists and split is list-splitting at a prefix, the inverse of concatenation—exactly the backward law of O'Neill's tensor-algebra free $\triangleleft$ -monoid. The branching case runs the same list-splitting path by path: Hedges' Kleene star is the

indexed generalisation of O’Neill’s tensor algebra, with branching supplied by the shapes and the backward law unchanged.

Status. Lemma 4.1 is proved above and verified computationally on 144,882 triples (t, u, z) over three containers, including a multi-shape bidirectional typechecker with a childless (answer-free) request and a branch-deleting graft; the $\triangleleft$ -monoid laws of C^* are formalised in Lean, while a Lean formalisation of Lemma 4.1 itself is a flagged target and not yet done. The universal property it serves is prior work; the new content is the isolation of the backward pass as path-confined.

4.3. Instance (III): factoring through the least element. The third instance is an axiom in the theory of arboreal categories, and we cite it.

Arboreal categories were introduced by Abramsky and Shah [2] as an axiomatic home for the game comonads of finite model theory—the Ehrenfeucht–Fraïssé, pebbling, and modal comonads, whose coalgebras encode plays of model-comparison games. A *path* in such a category is an object whose poset of subobjects (for a fixed proper factorisation system) is a finite linear order [7, Def. 4]; paths are the chains, the single branches. The original axioms required paths to be *connected*: every arrow from a path into a coproduct $\coprod_i Q_i$ of paths factors through a single summand [7, Def. 5].

As Jakl and Reggio observe [7, Ex. 8], this fails for the modal comonads: at the level of the underlying forest orders, coproducts glue trees at the root, and a path through such a gluing need not land in one summand. Their repair replaces connectedness with *tree-connectedness*.

Definition 4.4 (tree-connectedness, [7, Def. 11]). A path P is *tree-connected* if for every tree-diagram of paths $D : T \rightarrow \mathcal{A}$ and every arrow $f : P \rightarrow \text{colim } D$ there is an element $t \in T$ through which f factors via the colimit coprojection, and t is the *least* such element of T .

This is Schema 3.2 in guise (3): the factoring is through the *least* element of the tree T , the root of the relevant down-set. Jakl and Reggio show that tree-connectedness recovers all the properties that the connectedness axiom was used to prove, now including the modal comonads [7, Lem. 12], and that the resulting path functor is a Street fibration whose cartesian morphisms are exactly the *pathwise embeddings* [7, Thm. 23, Thm. 26]. We will use these two theorems in Section 5; for now the point is only that the repair *is* the move: replace an arbitrary-coproduct condition by factoring through a least tree element.

4.4. Instance (IV): the base is a forest. The fourth instance makes the move a property of a small category, and it is the one that becomes a theorem.

4.4.1. *Coalgebras of a polynomial comonad.*

Lemma 4.5. *Let $c = C$ be a polynomial comonad, viewed as a small category via Proposition 2.2. The category of Eilenberg–Moore coalgebras of the comonad $\llbracket c \rrbracket$ on $\mathbf{Set}$ is equivalent to the copresheaf category $[C, \mathbf{Set}]$, equivalently to the discrete opfibrations over C .*

Proof. A coalgebra is a map $\gamma : X \rightarrow \llbracket c \rrbracket X = \sum_{s \in S} X^{P_s}$. Write $\gamma(x) = (|x|, (x \cdot f)_{f \in P_{|x|}})$ with $|x| \in S = \text{ob } C$ the label of x and $x \cdot f \in X$. The counit law $\varepsilon \circ \gamma = \text{id}$ says $x \cdot \text{id}_{|x|} = x$. The coassociativity law $(\delta \triangleleft \text{id}) \circ \gamma = (\text{id} \triangleleft \gamma) \circ \gamma$ says, for $f : |x| \rightarrow a'$ in C and $g : a' \rightarrow a''$, that $|x \cdot f| = \text{cod}(f)$ and $x \cdot (g \circ f) = (x \cdot f) \cdot g$. These are exactly the data and laws of a functor $X : C \rightarrow \mathbf{Set}$ with $X(a) = \{x : |x| = a\}$ and $X(f)(x) = x \cdot f$: an object over each object, a chosen lift of each arrow out of it, respecting identities and composition—i.e. a discrete opfibration over C . Coalgebra morphisms are natural transformations. $\square$

4.4.2. *Arboreal = forest.*

Corollary 4.6. *If C is a forest category then every object of $[C, \mathbf{Set}]$ is a forest, by Lemma 3.1(1). For $C = [k]$ the finite linear order this equivalence is exact: $[C, \mathbf{Set}]$ is the category of forests of height $\leq k$; for $C = [k]^+$ the prefix trie, it is the trie-indexed forests. Hence the arboreal coalgebra categories of the Ehrenfeucht–Fraïssé and pebbling comonads are the copresheaf categories of their (forest) comonoids: for these game comonads, “arboreal = the base is a forest” is a theorem, not an analogy.*

Proof. A copresheaf $X : [k] \rightarrow \mathbf{Set}$ assigns a set X_n at each n and, for each arrow $n \rightarrow m$ (which exists iff $m \leq n$), an “ancestor” map $X_n \rightarrow X_m$. Grade each element by its label and order $x \leq y$ iff $|x| \leq |y|$ and $y \cdot |x| = x$; then every principal down-set $\{y \cdot m : m \leq |y|\}$ is a chain of length $|y| \leq k$ —a forest of height $\leq k$ by Lemma 3.1(1)—and depth-grading a forest inverts the construction. The trie case is identical with $[k]^+$ in place of $[k]$. $\square$

By Proposition 2.2 and a direct computation, the Ehrenfeucht–Fraïssé comonad $\mathbb{E}_k$ (nonempty sequences of length $\leq k$, counit last, comultiplication prefixes) has comonoid the linear order $[k]$, and the pebbling comonad $\mathbb{P}_k$ has comonoid the k -ary prefix trie $[k]^+$; the modal comonad is not polynomial over $\mathbf{Set}$, consistently with its being the instance that forced the tree-connectedness repair of Section 4.3. Thus Corollary 4.6 identifies the arboreal coalgebra categories of $\mathbb{E}_k$ and $\mathbb{P}_k$ —the forests of bounded height and the trie-indexed forests of Abramsky and Shah [2]—with the copresheaf categories of their polynomial comonoids. This is the bridge promised in the introduction.

Status and novelty. Lemma 4.5 is standard-adjacent (the familial / cofree-coalgebra story in Poly [8]) and Corollary 4.6’s forest identification is proved directly for $\mathbb{E}_k$ by Abramsky and Shah [2]; the increment claimed here is only the clean identification of the two, promoting “arboreality = forest” to a theorem and exhibiting it as the sharpest form of Schema 3.2. The coalgebra counts agree with forest counts in every tested case ($|X| = 2, 3$ and $k = 2, 3$).

5. READING THE FOUR AS ONE, AND AN OPEN PROBLEM

5.1. One move, four sharpenings. The four instances are the same move at four altitudes, each sharper than the last:

$$\underbrace{\text{root at each shape}}_{\text{(I) positions}} \longrightarrow \underbrace{\text{cut at the unique prefix}}_{\text{(II) free-monoid backward pass}} \longrightarrow \underbrace{\text{factor through the least element}}_{\text{(III) arboreal colimits}} \longrightarrow \underbrace{\text{the base}}_{\text{(IV) p}}$$

Instance (I) is a discipline on a position set; instance (II) a property of a backward map; instance (III) a property of colimit factorisations; instance (IV) a property of a small category. Lemma 3.1 is what they share: in each, the data below a node is a chain, so the operative map reads a unique root-path and ignores all branching. That the move appears independently in a frontier paper on arboreal categories [7] and in the backward pass of a free monoid, on structurally the same day and with neither borrowing from the other, is the evidence that it is a genuine normal form and not a coincidence of vocabulary.

5.2. Where the schema would next bite: an open problem. Instance (IV) identifies the arboreal coalgebra category with a copresheaf category *as a category*. The deeper question is whether the match persists *as a fibration*. Jakl and Reggio’s path functor $\mathbf{p} : \mathcal{A} \rightarrow \mathcal{T}$ is a Street fibration whose cartesian morphisms are the pathwise embeddings [7, Thm. 23, Thm. 26]; the Poly side has its own container fibration $\text{Cont}(\text{cod}) = \text{Fam}(\text{cod}^{\text{op}})$, a bifibration over the category of containers. Do these coincide?

Conjecture 5.1 (refined crown; *speculative*). *For $c = C$ a forest polynomial comonad, the restriction of the arboreal path fibration $\mathbf{p} : \mathcal{A} \rightarrow \mathcal{T}$ to the subcategory of paths is equivalent, as a fibration, to a Fam-type fibration built from C via $\text{Cont}(\text{cod})$ —up to the fibrewise opposite forced by the two sides computing dual (co-)intuitionistic predicate structures.*

We state Conjecture 5.1 as *speculative*, and we can say precisely how far the evidence reaches and where it stops. On the *unary, propositionally-truncated, single-path* fragment—one unary relation, subobjects rather than proof-relevant predicates, a chain rather than a branching tree—we have computed (by explicit enumeration at $C = [2], [3], [4]$ together with a k -independent derivation) the following, graded *computed*:

- the two fibres over a path $[n]$ are *opposite* posets— $\mathbf{p}$ gives colourings under $\subseteq$, $\text{Cont}(\text{cod})$ gives subobjects under $\supseteq$;
- the arboreal cartesian reindexing along a path inclusion is *preimage* $R \mapsto R \cap [m]$ —relation-reflection, substitution of predicates;
- this preimage is the *opcartesian* (pullback) leg ρ^* of the bifibration $\text{Cont}(\text{cod})$, *not* its cartesian leg $(\Sigma_\rho)^{\text{op}}$, which is direct image and differs from preimage already at the deepest singleton.

So on this fragment the arboreal path fibration is the fibrewise opposite of $\text{Cont}(\text{cod})$, and the two theories meet precisely on the pullback leg ρ^* : relation-reflection is ρ^* , not $(\Sigma_\rho)^{\text{op}}$. This is a usable dictionary—it tells

any application that needs to “pull a predicate back along a sub-trajectory” which leg of the container bifibration to use—but it is not the conjecture. Three gaps keep Conjecture 5.1 open, and we name them honestly:

- (1) *Path base only.* The single-shape container $\Phi([n]) = (1 \triangleleft [n])$ reaches only the paths $\mathcal{P} \subset \mathcal{T}$, not the branching trees; the full base needs multi-shape containers whose shapes are the branches, and the branch bookkeeping is not done.
- (2) *Unary only.* A k -ary relation on $[n]$ is a subset of $[n]^k$, which is not an object of $\text{Set}/[n]$; higher-arity relations are invisible to $\text{Cont}(\text{cod})$. The match is a statement about the unary fragment, not about all of finite model theory’s game comonads.
- (3) *Truncation.* The comparison is at the level of subobjects. The proof-relevant $\text{Cont}(\text{cod})$ -fibre $(\text{Set}/[n])^{\text{op}}$ is strictly richer than the arboreal $\text{Sub}([n])$, and a proof-relevant arboreal theory with witness-set colours is not standard.

We flag, finally, that two of the founding game-comonad references [2, 1] are cited here through the survey [7] and should be read directly before any formal submission; the load-bearing definitional content of this paper rests only on the deep-read sources.

6. CONCLUSION

One move—confine to a rooted tree, read the unique root-path—organises the root axiom of directed containers, the backward pass of the branching free monoid, the tree-connectedness repair of arboreal categories, and the forest condition on polynomial comonoids. Its single geometric content is that a forest has no branching below a node (Lemma 3.1), and in its sharpest form it is a theorem: the coalgebras of a polynomial comonad $c = C$ are the copresheaves on C , so arboreality is exactly the forest property of C (Theorem 1.1). This identifies the arboreal coalgebra categories of finite model theory with copresheaf categories of polynomial comonads—a bridge the two programmes had not drawn.

Two directions stand out. The first is to settle whether the four instances are specialisations of one *lemma*—a single statement, over a small category C with a factorisation system, equating “connectivity” with “ C a forest” and the operative map with the unique-prefix map—or whether they remain a family united only in spirit; instance (IV) supplies the vocabulary to attempt it. The second is Conjecture 5.1: on the unary truncated path fragment the arboreal path fibration is the fibrewise opposite of the container bifibration, meeting it on the pullback leg ρ^* ; whether this survives the passage to branching trees, higher arities, and proof-relevance is open, and the obstruction in each case is named above. Either direction would turn a structural coincidence into a structural theorem—which is, after all, the whole point of noticing that one keeps making the same move.

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